Label-Controlled Synthesis of Open and Closed Eye Images with a Conditional GAN

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Abstract

Eye-state classifiers for blink detection and driver-drowsiness monitoring need many labelled images of open and closed eyes. Synthetic data that can be generated on demand, in a chosen class, is a cheap way to enlarge or rebalance such datasets.

We train a conditional generative adversarial network (cGAN) on 41,649 grayscale 50×50 *eye images that produces an open or a closed eye on request. The class label is injected into* both *networks. The generator concatenates a learned 16-dimensional label embedding with its noise input. The discriminator receives the label as a learned* 50×50 *map stacked with the image as a second input channel, so it judges image–label* pairs. *After 100 epochs, flipping only the label on fixed noise switches the output between a closed lid and an open eye with a visible iris and highlight. Samples are diverse within each class and show no visible mode collapse. We describe the conditioning mechanism in detail, analyse the training dynamics, and propose quantitative checks for future work.*

1. Introduction

Why synthesise eyes? Open/closed eye datasets are slow to collect and label, and they are often imbalanced or narrow in their lighting and subjects. A generator that can be asked for "20 closed eyes" offers controllable augmentation and a way to probe what a classifier relies on.

Why a *conditional* GAN? A vanilla GAN [1] learns to map noise to realistic images but gives no control over which class it produces. A cGAN [3] feeds the class label y to both players. The generator learns $p(x \mid y)$, and the discriminator learns to reject images that are realistic but do not match their label.

Contributions.

- A DCGAN-style [6] cGAN for 50×50 eye images with label embeddings in the generator and a spatial label map in the discriminator.
- 20 synthetic images per class, plus fixed-noise progress grids that show the label controlling the output.
- An analysis of the training dynamics and a proposal for quantitative evaluation.

2. Related Work

GANs and DCGAN. Goodfellow *et al.* [1] introduced the adversarial minimax game; DCGAN [6] established the strided-convolution, batch-norm and ReLU/LeakyReLU design used here. Salimans *et al.* [7] proposed one-sided label smoothing to stabilise training.

Conditioning strategies. Mirza and Osindero [3] concatenated a one-hot label to the inputs of both networks. AC-GAN [5] instead asks the discriminator to also predict the class, and the projection discriminator [4] takes an inner product between a label embedding and the image features. We use learned embeddings concatenated at the input, a direct and interpretable extension of [3].

3. Method

3.1. Conditioning mechanism

Labels are $y \in \{0{=}\text{closed}, 1{=}\text{open}\}$. Both embedding tables are learned jointly with the GAN (Fig. 1).

Generator: label $\oplus$ noise. y is mapped by an embedding table to $e_G(y) \in \mathbb{R}^{16}$ and concatenated with noise $z \sim \mathcal{N}(0, I_{100})$. Because the 116-dimensional input enters the first layer, every later feature map depends on the label.

Discriminator: spatial label map. y is mapped to $e_D(y) \in \mathbb{R}^{2500}$, reshaped to a $1 \times 50 \times 50$ map and stacked with the image as a second channel. The discriminator thus sees the label at every pixel and can learn that, for example, a visible iris is inconsistent with "closed". A realistic image with the wrong label is rejected, which forces the generator to respect the condition.

Balanced generator updates. Generator steps sample their target labels uniformly at random, so both classes receive equal generator gradient regardless of the batch composition.

3.2. Architecture

Generator (3.71 M parameters). Linear(116→8192) is reshaped to $512 \times 4 \times 4$, followed by four stride-2 transposed convolutions ($512 \to 256 \to 128 \to 64 \to 1$, kernel 4) with batch norm and ReLU, giving 64×64. Adaptive average pooling resizes to 50×50, and a tanh maps to $[-1, 1]$.

Discriminator (0.66 M parameters). Three stride-2 convolutions ($2 \to 64 \to 128 \to 256$, kernel 4) with

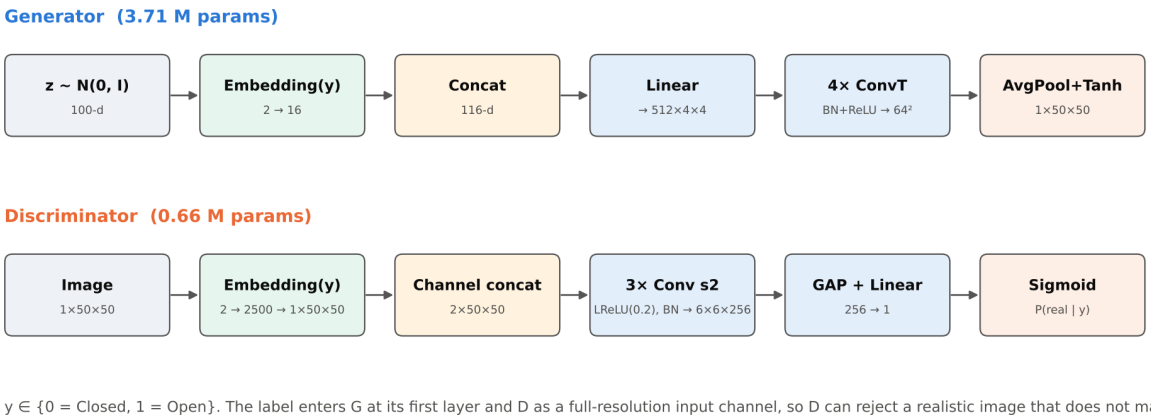


Figure 1. Generator (top) and discriminator (bottom). Green blocks are the learned label embeddings that carry the condition.

LeakyReLU(0.2), and batch norm on the last two, reduce the input to $256 \times 6 \times 6$. Global average pooling, a linear layer and a sigmoid give $P(\text{real} \mid y)$.

3.3. Objective

The players optimise the conditional minimax objective

$$\min_G \max_D \mathbb{E}_{x,y}[\log D(x \mid y)] + \mathbb{E}_{z,y}[\log(1 - D(G(z \mid y) \mid y))]\,. \tag{1}$$

It is implemented with binary cross-entropy. The discriminator is trained toward 0.9 on real pairs (one-sided label smoothing [7]) and 0 on generated pairs, and its loss is the mean of the two terms. The generator uses the non-saturating loss with target 1.

4. Results and Discussion

4.1. Data and training

The blink-detection dataset contains 19,753 closed-eye and 21,896 open-eye grayscale 50×50 images (41,649 in total), all used for training and normalised to $[-1, 1]$. Both networks use Adam (learning rate 2×10^{-4}, $\beta_1 = 0.5$, $\beta_2 = 0.999$), batch size 128 and 100 epochs. Convolutional weights are initialised from $\mathcal{N}(0, 0.02)$, and batch-norm scales from $\mathcal{N}(1, 0.02)$.

4.2. Training dynamics

Neither loss diverges or collapses (Fig. 3). The discriminator loss stays around 0.5–0.75 and the generator loss around 0.6–1.25. After epoch 50 there are recurring sharp dips in the generator loss, mirrored by spikes in the discriminator loss. This back-and-forth is normal adversarial dynamics rather than failure, but it means the last epoch is not necessarily the best; keeping checkpoints and selecting by a sample-quality metric would be safer.

4.3. Conditioning works: fixed-noise progress

Figure 2 shows the same 20 noise vectors at four checkpoints, with the label set to *closed* in the top row and *open* in the bottom row of each grid. At epoch 1 the outputs are texture. By epoch 20 eye structure appears, and by epoch 60 lids, lashes and irises are sharp. Within each column the noise is identical, so the difference between the two rows is caused by the label alone, and it is exactly the closed/open difference.

4.4. Generated samples

At inference we draw fresh noise and generate 20 images per class (Fig. 4). No real images are involved.

Class fidelity. Closed samples consistently show a lowered lid with lashes and no iris; open samples show a dark iris, often with a specular highlight.

Diversity. Within a class, samples vary in skin tone, lighting, eyebrow position and gaze. No two samples in Fig. 4 are near-duplicates, so there is no visible mode collapse.

Stable latent space. A given noise vector keeps a similar appearance from epoch 20 to 100 (Fig. 2): later training refines detail rather than reorganising the latent space.

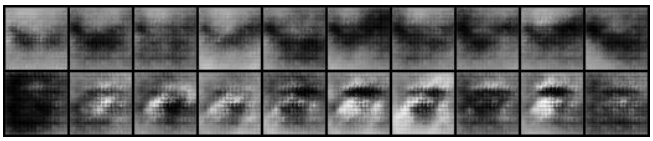
Quality limits. At 50×50 the images are soft. Closed-eye samples are more convincing than open ones, which sometimes show blurred or doubled irises.

5. Conclusion & Future Work

A cGAN with label embeddings at the generator input and a spatial label map in the discriminator produces open and closed eye images on demand, with clear label control and good diversity.

Limitations. The evaluation is visual only. The $64 \to 50$ average-pool resize slightly blurs outputs.

Next steps. (i) Measure conditioning accuracy directly by scoring generated images with an independently trained eye-state classifier; (ii) report per-class FID [2]; (iii) compare against AC-GAN [5] and projection-discriminator [4] con-



(a) Epoch 1



(b) Epoch 20



(c) Epoch 60

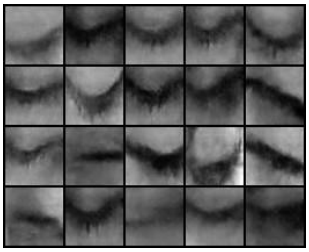


(d) Epoch 100

Figure 2. Fixed-noise samples during training. In each grid the top row uses label *closed* and the bottom row label *open*; each column shares one noise vector.



Figure 3. Per-epoch mean generator and discriminator losses.



(a) Closed ($y = 0$)



(b) Open ($y = 1$)

Figure 4. Twenty synthetic images per class from fresh noise vectors.

ditioning; and (iv) test whether adding synthetic eyes to a small real training set improves a downstream blink detector.

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References

[1] I. Goodfellow et al. Generative adversarial nets. In *Adv. Neural Inform. Process. Syst.*, 2014.

[2] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In *Adv. Neural Inform. Process. Syst.*, 2017.

[3] M. Mirza and S. Osindero. Conditional generative adversarial nets. *arXiv:1411.1784*, 2014.

[4] T. Miyato and M. Koyama. cGANs with projection discriminator. In *Int. Conf. Learn. Represent.*, 2018.

[5] A. Odena, C. Olah, and J. Shlens. Conditional image synthesis with auxiliary classifier GANs. In *Int. Conf. Mach. Learn.*, 2017.

[6] A. Radford, L. Metz, and S. Chintala. Unsupervised representation learning with deep convolutional generative adversarial networks. In *Int. Conf. Learn. Represent.*, 2016.

[7] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, and X. Chen. Improved techniques for training GANs. In *Adv. Neural Inform. Process. Syst.*, 2016.